applied sciences

MDPI

Review

A Systematic Literature Review on AI-Driven Predictive Maintenance and Fault Detection in Aircraft Systems

João Costa [1,*], José Torres Farinha [2], Hugo Raposo [2], Antonio J. Marques Cardoso [1], Alice Carmo [3], Paula Gonçalves [3] and João Farto [3]

[1] CISE—Electromechatronic Systems Research Centre, University of Beira Interior, Calçada Fonte do Lameiro, 6201-001 Covilhã, Portugal; ajmcardoso@ieee.org
[2] RCM2+—Research Centre for Asset Management and Systems Engineering, 3030-199 Coimbra, Portugal; tfarinha@isec.pt (J.T.F.); hugrap@gmail.com (H.R.)
[3] Portuguese Air Force Research Center, Granja do Marquês, Pêro Pinheiro, 2715-021 Sintra, Portugal; acduarte@emfa.gov.pt (A.C.); pagoncalves@emfa.gov.pt (P.G.)
* Correspondence: joaopaulopcosta1014@gmail.com

Abstract

The increasing availability of onboard sensors and digital monitoring platforms has enabled the continuous acquisition of operational and health-related data in aircraft systems. In parallel, advances in Big Data analytics and Artificial Intelligence (AI) have driven significant progress in Predictive Maintenance (PdM), enabling earlier fault detection and more reliable estimations of Remaining Useful Life (RUL). This systematic literature review examines recent developments in AI-driven PdM and fault detection applied to aircraft over the last years. A total of 20 studies were selected based on predefined inclusion criteria and analyzed with respect to research trends, application domains, algorithmic approaches, and expected outputs. The findings indicate a strong research emphasis on civil aviation supported by accessible operational datasets, whereas military aviation research prioritizes fleet readiness and mission continuity, often with limited data transparency. Deep learning approaches, particularly hybrid models combining convolutional and recurrent architectures, dominate recent prognostic methodologies, while optimization and Model-Based Systems Engineering (MBSE) frameworks support decision-making integration. Despite these advancements, the transition from experimental models to operational deployment remains constrained by data heterogeneity, model explainability requirements, and regulatory certification processes. This review highlights current progress and identifies gaps and research opportunities to accelerate the adoption of robust and scalable PdM solutions in aviation.

Keywords: predictive maintenance; aircraft health monitoring; fault diagnosis; Remaining Useful Life (RUL); prognostics and health management (PHM); military and civil aviation

Academic Editor: Andrea Prati

Received: 19 January 2026
Revised: 20 March 2026
Accepted: 26 March 2026
Published: 31 March 2026

1. Introduction

Maintenance has always been a cornerstone of aviation safety, reliability, and operational efficiency. From the earliest days of powered flight, ensuring aircraft airworthiness was crucial, and maintenance practices have evolved in parallel with technological advances, engineering knowledge, and operational demands. Initially, aviation maintenance was largely reactive, relying on corrective maintenance, which involves repairing or replacing components only after a failure has occurred. While this approach was feasible in the early 20th century, when aircraft systems were relatively simple, it posed significant safety risks, led to frequent operational disruptions, and generated high economic costs as aircraft became more complex and critical to transportation networks [1,2].

Notable historical accidents highlight how purely reactive maintenance could result in catastrophic outcomes, emphasizing the need for more proactive approaches. In response to these limitations, the aviation industry adopted preventive maintenance, which schedules inspections and replacements based on fixed intervals or usage metrics. This approach sought to reduce the likelihood of unexpected failures by servicing components before they reached the end of their anticipated life. While preventive maintenance significantly improved safety and reliability, it often resulted in unnecessary interventions, increased operational costs, and inefficient use of resources, since components were replaced regardless of their actual condition [3,4].

The emergence of Predictive Maintenance (PdM) marked a transformative shift in the philosophy of aircraft maintenance. Predictive maintenance leverages real-time monitoring, sensors, and advanced data analytics to evaluate the actual condition of aircraft components and systems. By forecasting potential failures, it allows maintenance to be scheduled more efficiently, reducing downtime, improving reliability, and optimizing resource use [5,6]. Prescriptive maintenance (RxM), a further evolution, combines predictive insights with decision-making algorithms to recommend optimal interventions, balancing safety, operational continuity, and cost considerations [7,8]. Together, these strategies illustrate a continuum in maintenance practices, evolving from reactive approaches toward increasingly proactive and intelligence-driven paradigms [9].

Technological advances have been pivotal in this evolution. The introduction of jet engines, fly-by-wire systems, and advanced avionics in the late 20th century increased both the complexity and criticality of maintenance tasks. Parallel innovations in sensors, data acquisition systems, and computational analytics enabled real-time monitoring of mechanical, electrical, and structural components. These technological developments facilitated the transition from schedule-driven to condition-driven maintenance, laying the foundation for predictive and prescriptive strategies [10]. Moreover, the advent of big data and Artificial Intelligence (AI) has enabled maintenance teams to analyze vast datasets from operational aircraft, supporting more accurate failure predictions and decision-making [11–13].

Comparing aviation to other industrial sectors underscores both the uniqueness and commonality of its maintenance evolution. In manufacturing, energy, and transportation industries, predictive and prescriptive maintenance approaches have been adopted to reduce downtime, extend equipment life, and optimize costs [14,15]. Aviation shares these goals but operates under stricter safety, regulatory, and reliability constraints, making the integration of predictive strategies both more challenging and more critical. Lessons learned from these sectors—such as sensor fusion, machine learning models, and digital twin technologies—have informed and accelerated the adoption of data-driven maintenance in aviation [16,17].

Despite technological progress, challenges persist in fully implementing predictive and prescriptive maintenance. Variability in aircraft types, operational conditions, regulatory requirements, and data management practices complicates the development of universally applicable maintenance models [18]. A particularly critical barrier is the validation and certification of predictive algorithms within aviation's regulatory ecosystem, where explainability and traceability are essential for safety compliance [1,19]. Additionally, data quality, sensor reliability, and the heterogeneity of fleet-level information remain key obstacles [5,7,10]. Scalability from experimental models to fleet-wide implementation also requires further research and standardization [15].

In summary, although predictive and prescriptive maintenance hold considerable promise for transforming aircraft maintenance practices, their effective translation from research to operational use depends on overcoming persistent challenges related to data maturity, model reliability, regulatory compliance, and integration within existing maintenance ecosystems. The present study aims to provide a systematic and analytically delimited review of research on AI-driven predictive maintenance in aircraft systems published between 2017 and 2025. More specifically, it examines how the field has evolved over this period, which aviation domains and system applications have been most actively investigated, which methodological approaches have been predominantly adopted, and what forms of maintenance intelligence these approaches are designed to generate, including fault detection, prognostics, Remaining Useful Life estimation, and fleet-readiness support. By synthesizing the evidence from the selected studies and situating contemporary data-driven developments within the broader historical evolution of maintenance paradigms, as illustrated in Figure 1, this review offers a structured perspective on the current state of the art while identifying the principal gaps and directions that may shape future research.

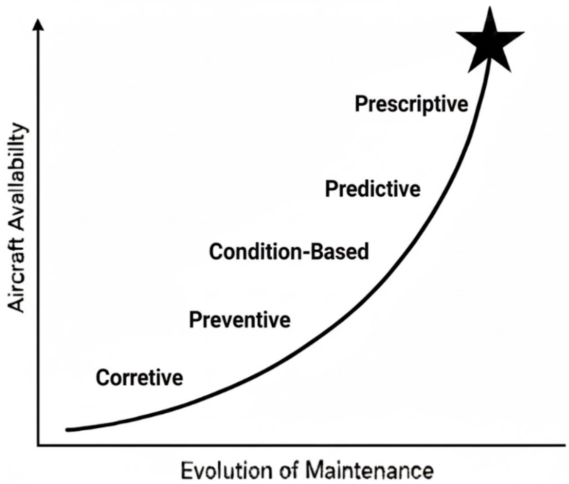

Figure 1. Timeline of evolution of maintenance.

2. Framework of Systematic Literature Review

A Systematic Literature Review (SLR) provides a rigorous and transparent methodology for identifying, evaluating, and synthesizing relevant research. In the context of PdM in aircraft, an SLR is essential to systematically assess trends, technological developments, and applications across different aircraft sectors. The framework of this SLR is designed to ensure reproducibility, minimize bias, and provide a structured understanding of the research landscape.

The framework consists of four main stages: (1) defining research questions, (2) identifying relevant literature, (3) selecting studies based on inclusion and exclusion criteria, and (4) analyzing and synthesizing findings. Each stage is described below.

2.1. Defining Research Questions

The review is guided by five specific research questions that focus on trends and patterns in PdM for aircraft:

- How has predictive maintenance research in aviation evolved over the last years?
- How are studies on PdM distributed across different academic publishers?
- How do predictive maintenance applications differ across aviation domains (e.g., commercial aviation, military, general aviation)?
- What categories of predictive maintenance algorithms are most commonly applied in aviation systems?

- What types of maintenance intelligence are generated by these models (e.g., fault diagnosis, prognostics, fleet readiness optimization)?

These research questions enable the review to encompass both the quantitative evolution and the qualitative characteristics of the literature. From a quantitative perspective, they support the analysis of publication trajectories over time, the concentration of scholarly output across specific publishers, and the distribution of studies among different aviation sectors. This allows a clearer understanding of how the field has developed, where research efforts are currently focused, and which operational contexts have received greater scientific and industrial attention.

From a qualitative perspective, the questions guide the examination of the technological strategies and methodological choices employed in aircraft predictive maintenance. Specifically, they facilitate the comparison between model-based, data-driven and hybrid PdM approaches, the identification of dominant algorithmic trends, and the evaluation of how these methods are tailored to address the safety, reliability and maintenance planning constraints inherent to aircraft systems.

Taken together, these quantitative and qualitative dimensions allow the review to construct a comprehensive characterization of the state of the art. This integrated perspective highlights not only the volume and distribution of research activity, but also the ways in which predictive maintenance is being conceptualized, operationalized and applied across aviation domains. In doing so, it supports the identification of existing research gaps, methodological patterns, and emerging opportunities that can inform future developments in the field.

2.2. Identifying Relevant Literature

This systematic literature review followed established evidence synthesis practices and was structured in accordance with the PRISMA framework (Preferred Reporting Items for Systematic Reviews and Meta-Analyses) to ensure transparency, reproducibility, and methodological rigor.

The literature search was conducted across four major scientific databases widely recognized for their coverage of aerospace engineering, reliability engineering, and artificial intelligence research:

- Scopus;
- Web of Science;
- IEEE Xplore;
- ScienceDirect.

These databases collectively index a large proportion of peer-reviewed publications in predictive maintenance, prognostics and health management (PHM), and aviation systems engineering. A structured search strategy was developed by combining predictive maintenance terminology with aviation-related keywords. Boolean operators were used to ensure broad coverage of relevant studies while maintaining domain specificity. The final search query was defined as:

("predictive maintenance" OR "prognostics and health management" OR "PHM" OR "condition-based maintenance") AND ("aircraft" OR "aviation" OR "aerospace" OR "UAV" OR "aircraft systems") AND ("military aviation" OR "defense aviation")

The search was restricted to publications between 2017 and 2025, a period characterized by rapid advances in artificial intelligence techniques and the increasing availability of large-scale operational datasets for predictive maintenance research.

The initial search returned 97 records. After removing duplicate entries, the remaining studies were subjected to a multi-stage screening process involving title review, abstract screening, and full-text eligibility assessment. Following this procedure, 20 studies were retained for qualitative analysis. The study selection process is summarized in the PRISMA flow diagram presented in Figure 2 and Supplementary Materials.

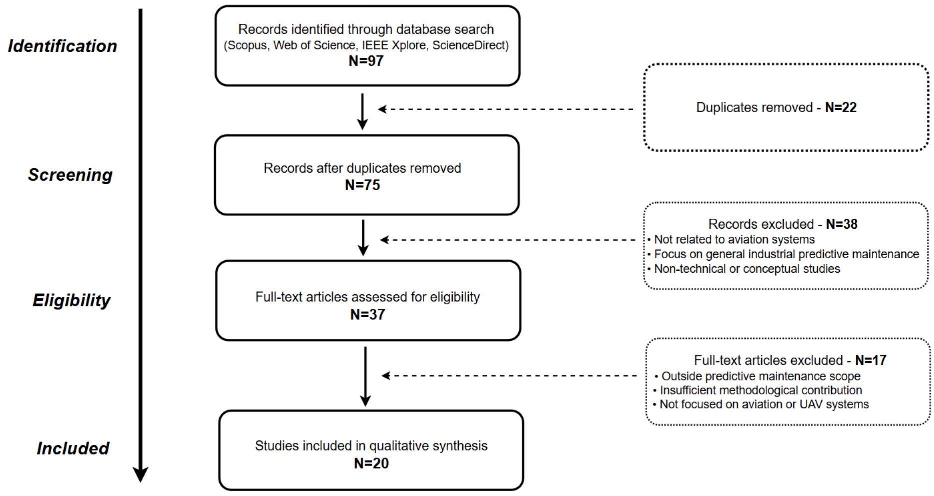

Figure 2. PRISMA flow diagram.

2.3. Study Selection: Inclusion and Exclusion Criteria

To ensure relevance and quality, studies were screened based on predefined criteria.

Inclusion criteria:

- Peer-reviewed journal articles, conference papers, and technical reports;
- Focused on aviation systems, aircraft components, aerospace systems, or UAV platforms;
- Were published between 2017 and 2025;
- Addressed predictive maintenance, prognostics and health management (PHM), or condition-based maintenance;
- English-language publications for clarity and accessibility.

Exclusion criteria:

- Studies unrelated to aviation maintenance or PdM.
- Articles without empirical evidence or methodological rigor.
- Duplicate publications or extended versions of previously published work.

2.4. Data Extraction, Analysis and Synthesis

For each selected study, relevant information was systematically extracted and organized to support comparative analysis. The extracted attributes included:

- Publication year;
- Publisher or journal;
- Application domain (e.g., civil aviation, military aviation, UAV systems);
- PdM algorithms or methodologies applied (e.g., machine learning models, statistical models, prognostics approaches);
- Type of output produced (e.g., fault detection, RUL prediction, fleet readiness).

The retrieved studies were analyzed using a combination of quantitative descriptive analysis and qualitative thematic interpretation. On the quantitative side, publication counts, sectoral distributions, and algorithm usage frequencies were examined to identify structural patterns in the research landscape. To support interpretability, a set of visualizations including temporal trend graphs, sector distribution maps, publisher contribution charts, and algorithm adoption plots was produced to provide a clear and systematic synthesis of the findings.

On the qualitative side, a thematic analysis was conducted to classify studies according to the underlying predictive maintenance paradigm (e.g., model-based, data-driven, hybrid) and to relate these methodological orientations to their operational application domains (commercial, military, or general aviation). This approach enabled the identification of recurrent modeling assumptions, data-dependent requirements, and operational constraints that shape how PdM techniques are adapted to aircraft systems, which are inherently safety-critical and maintenance-sensitive.

By applying this structured analytical framework, the review ensures methodological rigor, transparency, and reproducibility, while providing a coherent and integrated characterization of the state of the art. The combined quantitative–qualitative assessment makes it possible to discern research trends, sector-specific patterns, prevailing technological preferences, and existing gaps, thereby offering a robust basis for future research directions and the transition of predictive maintenance strategies into practical, aviation-ready applications.

3. Systematic Literature Review

This section presents the results of the systematic literature review (SLR) with the bibliometric analysis presented in Figure 3. To complement the systematic review, a bibliometric analysis was conducted in order to quantitatively characterize the research landscape of predictive maintenance in aviation. Bibliometric methods provide a structured approach for identifying trends, research activity patterns, and the evolution of scientific topics within a given field. The bibliometric dataset was derived from the same literature search protocol described in Section 2.

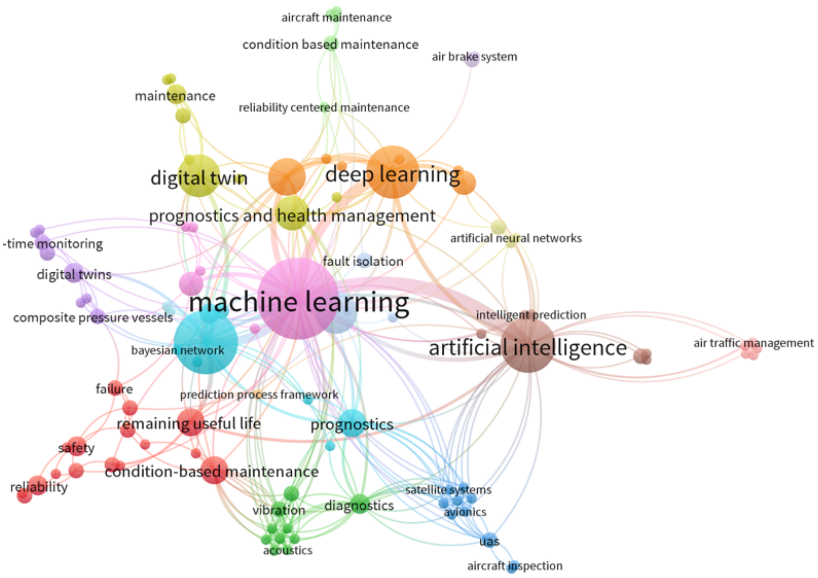

Figure 3. Bibliometric analysis.

The search was conducted across four major scientific databases—Scopus, Web of Science, IEEE Xplore, and ScienceDirect—which collectively provide broad coverage of peer-reviewed publications in aerospace engineering, reliability engineering, and artificial intelligence research. Bibliometric indicators extracted from the resulting corpus included:

- Application domain;
- Algorithmic approaches;
- Type of output produced.

These indicators were used to construct a descriptive overview of the field and to identify dominant research directions. From a methodological perspective, bibliometric analysis does not aim to evaluate the technical performance of individual algorithms but rather to map the structure and development of the research domain. By examining publication patterns, research topics, and methodological approaches, bibliometric analysis provides insight into how predictive maintenance research in aviation has evolved over time and which areas are currently receiving the greatest attention.

The relevance of this approach lies in its ability to complement qualitative literature analysis with quantitative evidence of research trends. In the context of predictive maintenance for aviation systems, bibliometric analysis helps identify:

- Emerging research topics;
- Dominant algorithmic paradigms;
- The evolution of predictive maintenance methodologies over time.

This combined approach, integrating systematic literature review with bibliometric analysis, enables a more comprehensive understanding of the current state of the field and supports the identification of future research directions. However, the bibliometric analysis also provides a structured quantitative overview of the reviewed corpus and establishes an objective basis for the interpretation of publication trends, methodological distribution, and thematic concentration within the field. Rather than serving as a purely descriptive complement, these indicators support the subsequent sections by enabling a more grounded examination of temporal evolution, publisher distribution, and methodological prominence. In the sections that follow, these quantitative patterns are explored through comparative and interpretative analysis, allowing the review to move from descriptive mapping toward a more critical assessment of how predictive maintenance research in aviation has developed in recent years.

3.1. Distribution by Database/Publisher

Figure 4 illustrates the distribution of the selected studies across the main academic databases and publishing platforms. The majority of the reviewed articles were retrieved from ScienceDirect, which accounts for the largest share, followed by IEEE Xplore and ResearchGate, while the remaining works were sourced from other platforms.

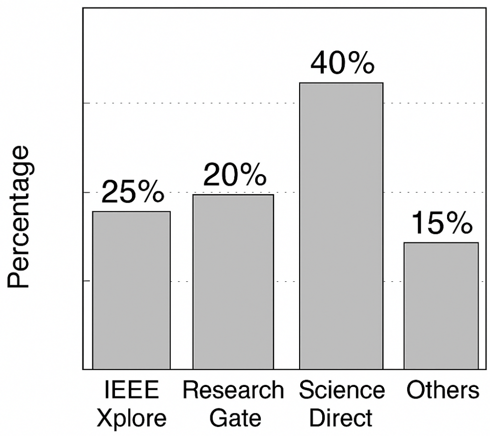

Figure 4. Distribution by Database/Publisher.

This distribution indicates that Elsevier and IEEE remain the primary publication ecosystems for predictive maintenance research in aviation, reflecting their strong presence in engineering, aerospace, and reliability-focused journals. ResearchGate appears mainly as a secondary discovery channel, frequently hosting preprints or author-uploaded manuscripts rather than original publication venues.

The concentration of studies in these databases suggests that predictive maintenance in aviation is predominantly disseminated through formal, peer-reviewed engineering outlets, where methodological rigor, algorithmic validation, and reproducibility are emphasized. Meanwhile, contributions found in the "Others" category (e.g., technical proceedings or institutional reports) provide practical context but generally exhibit lower methodological detail.

Overall, this distribution highlights a clear alignment between research maturity and publication venue: highly technical and algorithmic developments tend to appear in established scientific journals, whereas implementation-oriented documents are more often identified in non-journal repositories or organization-level publications.

3.2. Distribution by Publication Type

Figure 5 shows the distribution of the selected literature according to publication type. The majority of studies were published in peer-reviewed journal articles, indicating that research on predictive maintenance in aviation is being increasingly consolidated within established academic channels, where methodological rigor, validation protocols, and replicability standards are emphasized. Conference papers account for approximately 30 of the publications, reflecting the presence of ongoing exploratory research and the role of conferences as early dissemination venues for emerging techniques and prototype implementations.

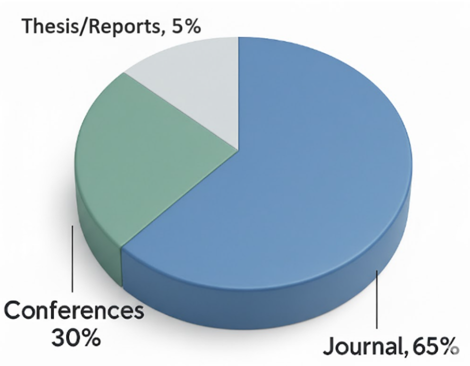

Figure 5. Distribution by publication types.

A smaller portion of the literature corresponds to theses and technical reports, which typically provide more extensive methodological detail or application-specific insights, often linked to industrial or defense-oriented contexts. Although these documents are valuable for understanding practical deployment conditions and real operational environments, they are less frequently indexed in mainstream databases and therefore appear less prominently in the published academic record.

Taken together, this distribution suggests that predictive maintenance for aircraft constitutes a maturing research domain, where core methodologies are increasingly formalized in journal literature, while innovation and system-level experimentation continue to be introduced through conference publications and institutional or technical reports. The balance observed reinforces the dual nature of the field: it is simultaneously advancing as a scientific discipline and being translated into operational practice within aviation environments.

https://doi.org/10.3390/app16073381

4. Answers to the Research Questions (RQ)

4.1. RQ1—How Has Predictive Maintenance Research in Aviation Evolved over the Last Years

Following the studies presented in Table 1, Figure 6 illustrates a progressive increase in research activity on predictive maintenance in aviation over the past five years. The period between 2017 and 2020 is characterized by a relatively low and stable volume of publications, corresponding to what may be regarded as a formative stage of the field. During this phase, scholarly contributions focused primarily on establishing conceptual foundations and on providing initial reviews and sectoral mappings of PdM adoption in aerospace systems [16,20].

Table 1. Summary of studies on predictive maintenance and diagnostic methods in aviation and related sectors.

Ref.	Field	Equipment	Algorithm	Output	Publ.
[21]	Civil	Aircraft Engine	Grouped Conv. Denoising Autoencoder	Fault Detection	Elsevier
[20]	Industrial	Aeronautical Sector	Conceptual Framework	Sectoral Adoption & Challenges	Elsevier
[22]	Civil	Aircraft Subsystems	Statistical Analysis of Flight Data	Early Fault Detection	SIMTech
[23]	Civil	Multi-Aircraft Fleet	Machine Learning	Large-Scale PdM Implementation	J. Propulsion Tech.
[24]	Civil	Aviation Ops	AI Framework	PdM Integration in Operations	JAIGS
[25]	UAV	Propeller System	Deep CNN	Damage Detection & Classification	Preprint
[26]	Civil	Aircraft	Data-Driven Model + RUL Estimation	Health Monitoring	Thesis
[27]	Military	Fleet Maintenance Scheduling	MILP + Heuristics Optimization	Readiness/Co-Planning	Elsevier
[28]	Military	PdM Process Integration	MBSE + Simulink	Scheduling Framework	IEEE
[6]	(Civil/Military)	Aviation Systems Safety	Decision-Support Framework	Prescriptive Maintenance Policy	MDPI
[29]	Military	Fleet Scheduling	Deep Learning	Readiness Optimization	Springer
[30]	General	Review Study	Generative AI	Data Augmentation/SLR	
[31]	Civil	Flight Control Actuator	SVM	Fault Detection	MDPI
[32]	General	Subsystem Network	Graph Convolutional Network (GCN)	System Fault Detection	ACM CIBDA
[33]	Civil	Aircraft Systems	Hybrid AE–CNN–BiGRU Model	RUL Prediction/Prognostics	Springer
[16]	Aerospace	Engine Subsystems	CNN-Based Feature Extraction	Component Fault Detection	Conf. Paper
[34]	Civil	Generic Aircraft	CNN vs. RNN/LSTM	Model Performance Comparison	ICOM Conf.
[35]	Civil	Fleet (Anomaly Detection)	Isolation/Anomaly ML Methods	Health Monitoring	MDPI
[36]	Civil	Multiple Systems	Digital Twin + RUL Models	Predictive Maintenance via DT	IEEE
[37]	Civil	Multi-Component System	RUL + PM Optimization	Spare-Constrained Strategy	Elsevier

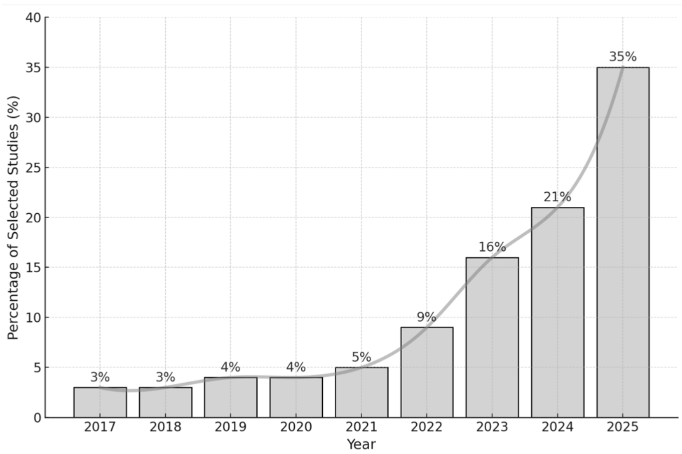

Figure 6. Annual growth of publications by years.

A notable expansion in publication volume and methodological diversity is observed from 2021 onwards, coinciding with the maturation of data acquisition infrastructures and the wider adoption of machine learning in aerospace maintenance contexts. The literature from 2023 reflects a clear shift from isolated component-level studies towards integrated maintenance frameworks. Examples include the joint optimization of flight operations and maintenance scheduling under uncertainty [27], actuator-level diagnostic schemes based on support vector machines [31], hybrid deep learning architectures that combine autoencoders, CNNs, and BiGRU networks for degradation trajectory prediction. Concurrently, contributions employing Model-Based Systems Engineering (MBSE) frameworks sought to address the integration of PdM outputs into operational decision processes [28].

Since 2024, research has shown a significant increase in the number of publications, with particular emphasis on AI scalability and methodological comparison. Key contributions include a review of generative AI for aviation data augmentation [30], comparative evaluations of CNN versus LSTM architectures for fault prediction [34], UAV in-flight propeller damage detection frameworks [25], and broader analyses of AI-enabled PdM strategies across the aviation ecosystem [24].

The most recent contributions, published in 2025, suggest a continued transition towards model integration and operational deployment. These include the use of graph convolutional networks for topology-aware degradation modelling [32], the formulation of decision-making frameworks for aviation safety management informed by predictive insights [6], and fleet-level maintenance scheduling methods enhanced with deep learning-based chance constraints [29].

Taken together, the temporal distribution of studies indicates a clear trajectory of maturation, moving from foundational algorithm design toward operational readiness and system-level adoption. PdM in aviation is thus evolving from an experimental research domain into a strategic capability for improving aircraft availability, cost efficiency, and mission assurance.

The temporal evolution presented in Figure 5 reveals a marked and sustained acceleration in aviation PdM research. While publication volume remained relatively modest and stable between 2017 and 2020, the field began to gain momentum from 2021 onward, followed by a sharp increase in 2022 and a substantial surge in 2024–2025. This transition reflects a shift from incremental experimentation toward a phase of methodological consolidation and practical relevance.

The pronounced growth observed since 2024 suggests that the scientific community is increasingly focused on improving scalability, benchmarking modelling approaches, and validating PdM frameworks under operational conditions. The research landscape is no longer centred on isolated algorithmic proposals but is moving toward integrated, system-level solutions capable of supporting real-world decision-making.

In conclusion for RQ1, the literature demonstrates that predictive maintenance in aviation has entered a mature and strategically significant stage. The field has evolved from early exploratory work to comprehensive architectures and validation studies that directly contribute to fleet reliability, cost optimisation and mission assurance. The sustained upward trend confirms both the relevance and the long-term momentum of this research domain.

4.2. RQ2—Distribution of Studies by Publishers' Fields

The distribution of studies across publishers highlights the multidisciplinary scope of PdM in aviation.

Elsevier outlets dominate several key contributions, such as readiness optimization in *Reliability Engineering and System Safety* [27] and convolutional denoising approaches in the *Chinese Journal of Aeronautics* [21].

MDPI journals provide significant coverage through surveys and applied research, including a defense-focused review in *Sensors* [20], actuator diagnostics in *Machines* [30], and prescriptive decision frameworks in *Applied Sciences* [6].

IEEE venues, such as the Aerospace Conference, bridge research and application by embedding PdM into MBSE environments [28]. Springer outlets showcase methodological depth, with hybrid deep learning architectures in *Neural Computing and Applications* [33] and fleet scheduling for readiness in *Autonomous Intelligent Systems* [29]. ACM provides structural advances with graph convolutional models [32], while additional contributions are distributed across conferences [26], university theses [26], and institutional reports [22].

This distribution confirms that PdM is situated at the intersection of reliability engineering, machine learning, systems engineering, and defense research, each domain contributing a complementary perspective to the field.

4.3. RQ3—How Do Predictive Maintenance Applications Differ Across Aviation Domains?

Figure 7 presents the distribution of the reviewed studies across different aviation domains, including civil aviation, general aviation, military aviation, aerospace, and unmanned aerial vehicles (UAVs). The results indicate that civil aviation constitutes the primary focus of research efforts, with eight studies addressing predictive maintenance in the commercial transport sector. These works commonly investigate aero-engine health monitoring [21], actuator-level diagnostics [31], hybrid prognostic models for commercial fleets [33], and airline-level case studies [26]. The predominance of civil aviation research reflects the availability of structured operational data, standardized maintenance records, and strong economic incentives associated with fleet reliability and reduced operational downtime.

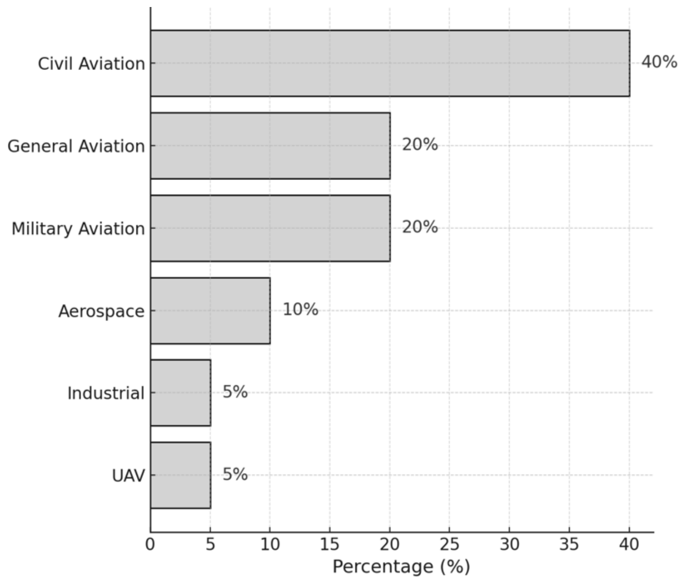

Figure 7. Distribution of Studies by Different Fields.

A substantial portion of the literature also concentrates on military aviation, which accounts for four studies. Research in this domain is generally oriented toward mission readiness and operational assurance, including stochastic flight and maintenance scheduling under uncertainty [27], the integration of predictive analytics into Model-Based Systems Engineering (MBSE) frameworks [28], decision-support systems for safety and maintenance management [6], and fleet-level optimization approaches employing advanced learning and constraint modeling [29].

UAV applications, though limited in number (one study in the current corpus), constitute an emerging research frontier. Work in this domain demonstrates real-time onboard diagnostic capabilities, particularly in propeller damage detection and severity estimation using inertial sensing data under dynamic flight conditions [25].

A further subset of studies contributes general or methodological frameworks, such as sectoral case analyses [20], CNN-based aerospace applications [16], and systematic reviews addressing generative AI or model-based diagnostics [30,35].

As a result, the current literature exhibits a structural asymmetry: civil aviation research dominates the academic landscape and drives algorithmic innovation, while military aviation demonstrates stronger operational deployment of predictive maintenance systems but remains underrepresented in peer-reviewed publications. This asymmetry also reflects the distinct institutional and operational environments in which the two aviation sectors operate. Civil aviation is governed by highly standardized regulatory frameworks that encourage data transparency and the publication of technical research. Military aviation, by contrast, operates under security-driven environments where operational data are frequently restricted.

Finally, these findings indicate that predictive maintenance research in aviation is currently evolving along two partially parallel trajectories. Civil aviation continues to advance methodological developments in predictive algorithms, while military aviation emphasizes the integration of predictive maintenance into operational readiness management. Bridging these two domains may represent an important direction for future research, particularly through the development of secure data-sharing frameworks and domain adaptation techniques capable of transferring predictive models across operational contexts.

4.4. RQ4—What Categories of Predictive Maintenance Algorithms Are Most Commonly Applied in Aviation Systems?

The methodological spectrum identified in the reviewed literature is diverse and reflects different levels of analytical maturity within aviation predictive maintenance. At the methodological level, the literature is dominated by learning-based and model-based approaches designed to support fault diagnosis, prognostics, and health-state estimation. Their applicability, however, depends not only on predictive performance but also on operational constraints related to data availability, computational efficiency, certification requirements, and explainability.

Deep temporal models, including LSTM, GRU and BiGRU, are widely used for multivariate prognostics [33,34]. Convolutional approaches remain prominent, with grouped denoising convolutional autoencoders applied to engine fault detection [21] and CNN-based frameworks proposed for aerospace systems [16]. These models demonstrate superior performance when large volumes of high-frequency sensor data are available, particularly in aircraft engines and complex subsystems where degradation patterns evolve over long operational cycles. However, these models typically require extensive training datasets and significant computational resources, which may limit their deployment in real-time onboard systems.

Classical machine learning retains utility, with support vector machines applied to actuator fault detection [31]. More recently, structural methods, such as graph convolutional networks, have emerged to represent subsystem interdependencies [32]. This approach remains relevant in contexts where datasets are smaller or partially labeled, offering lower computational complexity and improved interpretability. These characteristics make them particularly suitable for applications where regulatory transparency and model traceability are critical.

Beyond predictive modelling itself, the reviewed literature also includes a distinct group of operational integration approaches, particularly optimization-based methods and MBSE frameworks [27], which extend predictive maintenance from diagnostic inference to actionable scheduling and fleet-level decision support [28]. Unlike learning algorithms, these approaches do not primarily generate maintenance intelligence directly; rather, they structure, translate, and operationalize predictive outputs within broader maintenance and mission-planning processes [36].

Accordingly, Figure 8 should be interpreted as representing the principal methodological and integration approaches identified in the literature, rather than a single flat category of algorithms. This distinction is important because it highlights that aviation predictive maintenance operates across multiple analytical layers, ranging from component-level predictive modelling to system-level operational integration.

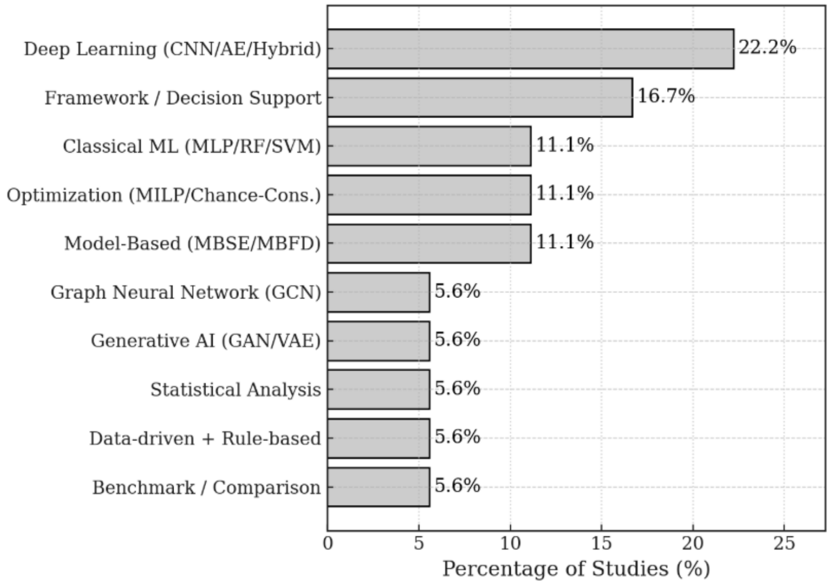

Figure 8. Frequently used algorithms and methods.

4.5. RQ5—What Types of Maintenance Intelligence Are Generated by These Models?

Having distinguished between methodological approaches and operational integration frameworks, the analysis now turns to the output dimension of predictive maintenance systems. This level of analysis concerns the specific forms of maintenance intelligence generated by the reviewed approaches, thereby complementing the preceding methodological classification with a clearer understanding of their functional purpose in aviation contexts.

The outputs identified in the reviewed studies can be grouped into four categories, as illustrated in Figure 9. The first and most frequent is fault detection and classification, applied to engines [21], actuators [13], UAV propellers [25], and complex subsystems via graph models [32].

The second is prognostics, particularly Remaining Useful Life (RUL) estimation, enabled by recurrent and hybrid deep learning models [33,34].

The third category is fleet readiness and availability, where military-focused studies explicitly optimize mission-capable rates through scheduling and resource allocation [27–29].

The fourth group includes frameworks and reviews, where the primary output is methodological: taxonomies, evaluation pipelines, or adoption-oriented frameworks [6,20,30,37].

Civil aviation research emphasizes technical outputs, such as RUL and fault detection, while military aviation emphasizes operational readiness. UAV studies add a practical dimension by validating lightweight in-flight monitoring approaches.

Together, these outputs illustrate both the complementarity and the current gap between academic research, which often targets component-level health and defense applications, which require system-level readiness outcomes.

However, beyond simple categorization, a structural relationship emerges between the algorithmic paradigm employed and the operational outputs generated by predictive maintenance models. In aviation applications, algorithm selection is closely linked to the type of maintenance intelligence required.

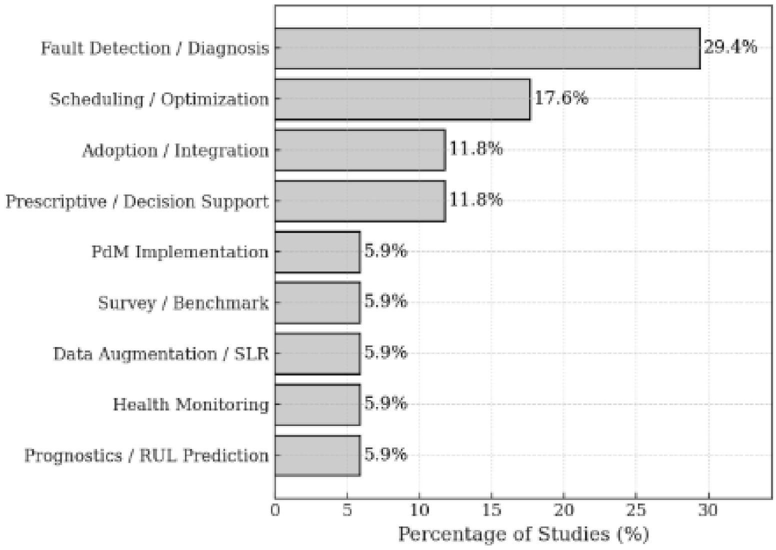

Figure 9. Distribution of Studies by PdM Output Parameters.

Deep learning architectures, particularly recurrent neural network models, are predominantly associated with prognostic tasks, such as Remaining Useful Life (RUL) estimation. These models are capable of capturing long-term temporal dependencies within multivariate sensor data streams and are therefore well suited to degradation trajectory modelling [33,34].

In contrast, classical machine learning algorithms, including Support Vector Machines (SVMs) and Random Forests, are more commonly applied to fault detection and anomaly classification, where the objective is to distinguish abnormal operational patterns within relatively shorter time windows [31,32].

Optimization-based approaches and Model-Based Systems Engineering (MBSE) frameworks typically operate at a higher system abstraction level. Rather than directly producing diagnostic predictions, they transform prognostic information into maintenance scheduling strategies and fleet readiness optimization decisions [28].

This layered relationship highlights the multi-level architecture of predictive maintenance systems in aviation, where data-driven diagnostics feed into operational decision-making frameworks.

4.6. Comparative Assessment of Predictive Maintenance Approaches

While the preceding sections describe the main research trends, application domains, algorithmic families, and maintenance outputs identified in the reviewed literature, a comparative perspective is also necessary to better understand the relative position of these approaches in aviation predictive maintenance. In particular, beyond their analytical capabilities, the practical relevance of different methods depends on factors such as data requirements, interpretability, and their current level of implementation in operational environments.

To provide this additional perspective, Table 2 presents a concise comparison of the principal methodological approaches identified in the reviewed studies, highlighting their expected performance, dependence on data availability, level of interpretability, and degree of real-world implementation. This comparison helps clarify the trade-offs that currently shape the adoption of predictive maintenance technologies in aviation.

Table 2. Comparative assessment of the main predictive maintenance approaches in aviation.

Approach	Performance	Requirements	Interpretability	Implementation
Classical ML models	High in structured fault detection tasks	Moderate	High	Moderate
Deep learning	High especially for complex time-series prognostics	High	Low	Low to moderate
Model-based approaches	high when system physics are well defined	Moderate	High	Moderate
MBSE frameworks	High for maintenance planning and decision support	Moderate to high	High	Moderate
Generative AI	Promising as support tools for model training	High	Low	Low

As shown in Table 2, the reviewed approaches differ not only in analytical strength but also in their suitability for aviation deployment. Deep learning methods generally offer the strongest predictive capability, particularly in prognostic tasks involving large volumes of multivariate sensor data, but their operational uptake remains constrained by high data demands and limited interpretability. By contrast, classical machine learning and model-based approaches appear more compatible with validation and certification requirements, since they are typically easier to interpret and integrate into controlled maintenance settings.

Optimization and MBSE-based frameworks occupy a complementary position, as their main contribution lies not in fault prediction itself but in translating maintenance intelligence into operational planning and decision support. Finally, generative AI approaches remain at an early stage of maturity, with potential value in addressing data scarcity, although their direct implementation in aviation maintenance practice is still limited. Overall, the comparison suggests that the technological maturity of predictive maintenance in aviation remains uneven, with methodological performance often advancing faster than real-world implementation.

5. Discussion and Conclusions

5.1. Data Heterogeneity and Certification Challenges

The results of the reviewed corpus indicate that, despite the increasing methodological sophistication of predictive maintenance research in aviation, the transition from analytical development to operational deployment remains constrained by recurring implementation barriers. Across the selected studies, these barriers emerge most consistently in relation to heterogeneous data environments, limited transparency of advanced models, and the difficulty of aligning AI-based predictions with certification-oriented aviation requirements. Both issues significantly affect the ability to integrate advanced machine learning models into real-world aviation maintenance systems.

Aircraft health monitoring systems generate large volumes of operational data originating from multiple sources, including onboard sensors, maintenance logs, flight operational parameters, and environmental measurements. These data streams are inherently heterogeneous, meaning they differ in structure, sampling frequency, format, and semantic interpretation. For example, vibration signals from rotating components may be recorded at high sampling rates, while flight parameters such as altitude or temperature are collected at lower frequencies. In addition, maintenance records and inspection reports are often stored as unstructured textual information, making them difficult to integrate directly with sensor data. This heterogeneity creates several challenges for predictive maintenance models. Machine learning algorithms typically require consistent, well-structured datasets, yet aviation data often contain missing values, inconsistent time synchronization, and varying levels of data quality across aircraft fleets.

Several approaches have been proposed to mitigate these challenges. Data fusion techniques allow heterogeneous sensor signals to be combined into unified representations, while feature engineering and time-alignment methods, which help synchronize measurements collected at different frequencies, have shown promise in integrating heterogeneous aviation data sources into predictive maintenance frameworks. Despite these advances, the effective management of heterogeneous data remains a fundamental challenge for large-scale predictive maintenance systems.

Beyond data challenges, predictive maintenance systems must also comply with strict certification requirements imposed by aviation regulatory authorities. Any system that influences maintenance decisions must demonstrate traceability, reliability and explainability to ensure compliance with airworthiness regulations.

This requirement creates difficulties for many modern machine learning models. Deep neural networks, for example, are often described as black-box models because their internal decision processes are difficult to interpret. While these models may achieve high predictive accuracy, the lack of transparency makes it challenging for engineers and regulators to understand how specific predictions are generated. To address this limitation, recent research has increasingly focused on Explainable Artificial Intelligence (XAI) techniques. The objective of XAI is to make machine learning models more transparent by providing human-understandable explanations for their predictions. In the context of predictive maintenance, XAI methods help engineers identify which sensor measurements or operational variables contribute most strongly to a predicted fault or degradation event. This transparency improves trust in AI systems and facilitates their integration into safety-critical environments such as aviation.

Among the most widely used explainability techniques is SHAP (SHapley Additive exPlanations). SHAP is based on concepts from cooperative game theory and quantifies the contribution of each input variable to a model's prediction. In practical terms, SHAP values allow engineers to determine how different sensor measurements, such as temperature, vibration, or pressure, influence the predicted health state of an aircraft component. This level of interpretability provides valuable diagnostic insight and allows maintenance engineers to validate model predictions against known physical behaviour.

Integrating explainability mechanisms into predictive maintenance models represents a critical step toward regulatory acceptance and certification of AI-based maintenance decision support systems. By improving transparency and providing traceable reasoning processes, XAI techniques can help bridge the gap between advanced machine learning models and the stringent safety requirements of the aviation industry.

Ultimately, the combination of improved data integration strategies and explainable AI methodologies will be essential for enabling the reliable and certifiable deployment of predictive maintenance technologies in next-generation aviation systems.

5.2. Conclusions

The systematic review conducted on the selected corpus of twenty studies from 2017 to 2025 offers a comprehensive perspective on the state of predictive maintenance (PdM) in aviation. The analysis of trends, publication outlets, application domains, algorithms, and output parameters reveals both the maturity and the fragmentation of the field. Over the past few years, research activity has increased steadily, reaching a peak in 2025. This growth trajectory demonstrates that PdM in aviation has moved from a niche interest to a consolidated area of investigation and practice.

The role of publishers and dissemination channels is especially revealing. Academic journals, such as those from MDPI, Elsevier, IEEE and Springer, remain central for methodological development and algorithm validation, particularly when leveraging benchmark datasets like C-MAPSS and N-CMAPSS. At the same time, defense and government sources provide crucial insights into the real-world adoption of PdM in aviation. This dual system of knowledge dissemination underscores the contrast between the civil and defense ecosystems: while academia focuses on openness, reproducibility, and methodological refinement, defense organizations emphasize practical readiness and mission assurance, often communicated through less formalized but highly impactful reports.

The distribution of reviewed studies across application domains reflects a dominance of civil aviation, primarily driven by the availability of open datasets that allow for benchmarking of algorithms and reproducible experiments. Research on military aviation, though less represented in peer-reviewed outlets, is nonetheless prominent in operational reports and case studies. UAV-related PdM is emerging as a growing area, reflecting broader technological trends where unmanned platforms are increasingly relied upon for inspection, logistics, and tactical missions. This distribution highlights a significant imbalance: civil research excels in algorithmic innovation, while military practice demonstrates large-scale operational implementation, yet the two rarely intersect in terms of published scientific outputs.

From an algorithmic perspective, the reviewed literature indicates the clear predominance of recurrent neural networks, especially LSTM-based architectures, which remain the standard for handling time-series degradation data. Hybrid models that combine physics-based knowledge with machine learning are gaining attention for their ability to deliver more interpretable and trustworthy outputs, a crucial consideration for mission-critical applications. Classical machine learning methods, while less frequent, still serve important roles in contexts with limited or sensitive datasets, often relevant for military environments. Reinforcement learning, although present in a smaller number of studies, points toward a future where maintenance scheduling and resource allocation can be optimized dynamically. The overall picture is of a field that is methodologically rich, with different approaches serving complementary roles depending on the availability of data, the intended application and the operational constraints.

The outputs produced by the studies are likewise heterogeneous. Civil-focused research overwhelmingly prioritizes the prediction of Remaining Useful Life, an output well aligned with the datasets most frequently used. In contrast, military studies and reports shift the emphasis toward fleet readiness and mission availability, reflecting the operational imperatives of defense organizations. Fault detection and classification, though less prevalent, emerge as particularly relevant for UAV platforms where vibration monitoring and anomaly detection play critical roles. Survey and review papers, meanwhile, provide broader methodological syntheses, situating the field within larger trends in prognostics and health management. This divergence of output priorities again illustrates the civil–military divide: while civil research optimizes at the component level, military practice is oriented toward system-wide availability and readiness.

The synthesis of these findings reveals both achievements and gaps. PdM in aviation has clearly matured, but its knowledge ecosystem is fragmented between academic innovation and military operationalization. Civil researchers benefit from accessible datasets, yet this reliance risks overspecialization on simulated or narrowly scoped data. Military practitioners demonstrate the viability of PdM at fleet scale, but much of this knowledge remains inaccessible to the scientific community due to security restrictions. Bridging these two spheres remains one of the central challenges for the coming years. The development of secure data-sharing frameworks, domain adaptation methods to transfer civil-trained models into military environments, and hybrid approaches that integrate expert knowledge with machine learning may offer viable paths forward.

This review also highlights several limitations. By focusing on twenty representative studies, the analysis provides depth and clarity but may not capture the full breadth of approaches across the literature. The reliance on published and publicly accessible sources inevitably excludes classified projects, which may already be advancing beyond what is visible in the public domain. These limitations should be borne in mind when interpreting the findings.

In conclusion, predictive maintenance in aviation has entered a phase of consolidation and operational significance. Civil research, enabled by open datasets, continues to drive algorithmic advances, especially in RUL prediction. Military aviation, less visible in the academic record, is nonetheless at the forefront of implementation, with large-scale systems, influencing readiness outcomes. UAVs represent the next frontier, expanding the reach of PdM to new classes of aircraft and operational contexts. The diversity of algorithms and outputs reflects a field that is both methodologically sophisticated and strategically important.

Finally, this review highlights a significant divergence between civil and military aviation that extends beyond technological aspects and is deeply rooted in institutional and regulatory structures.

Civil aviation operates within a highly standardized regulatory ecosystem governed by international authorities. These institutions enforce strict airworthiness certification processes, standardized maintenance procedures and transparent reporting mechanisms. As a result, airlines, aircraft manufacturers and research institutions frequently publish technical findings, enabling the development of publicly accessible datasets and benchmarking frameworks.

Military aviation, in contrast, operates within security-driven institutional environments where operational data, maintenance records, and system architectures are frequently classified due to national defense considerations. Consequently, predictive maintenance initiatives within military organizations often focus on operational deployment rather than academic publication.

Furthermore, the optimization objectives differ significantly between the two domains. Civil aviation predictive maintenance primarily aims to reduce maintenance costs and improve aircraft availability through component-level prognostics. Military aviation, however, prioritizes mission readiness and operational capability, requiring maintenance strategies that integrate scheduling, logistics, and mission planning. These institutional and operational differences explain the observed asymmetry in the literature, where civil aviation dominates algorithmic development while military aviation demonstrates stronger integration of predictive maintenance within fleet management and operational decision frameworks.

5.3. Future Research Directions

The findings of this review point to several strategic directions for advancing predictive maintenance research in aviation.

First, the integration of predictive maintenance models with digital twin architectures represents a promising pathway toward real-time synchronization between physical aircraft systems and their virtual counterparts, enabling continuous health assessment and more adaptive maintenance strategies.

Second, the development of explainable and certifiable artificial intelligence models will be essential to ensure regulatory compliance and to support the deployment of predictive maintenance solutions in safety-critical aviation environments, where transparency, traceability, and reliability are mandatory.

Third, the establishment of standardized aviation predictive maintenance datasets would substantially improve benchmarking capabilities and enable more consistent methodological comparisons across studies. Such datasets would also support reproducibility and accelerate algorithm development within the research community.

Fourth, future research should address the current gap between component-level prognostics and fleet-level operational decision-making. Bridging this gap would allow predictive insights to directly inform maintenance planning, logistics coordination, and fleet readiness management, thereby enhancing the operational value of predictive maintenance systems.

Finally, stronger collaboration between academia, aircraft manufacturers, airlines, and defense organizations will be necessary to overcome existing data accessibility barriers and facilitate the transition from experimental predictive models to fully operational predictive maintenance frameworks within aviation systems.

Supplementary Materials: The following supporting information can be downloaded at: https://www.mdpi.com/article/10.3390/app16073381/s1, Document S1: PRISMA 2020 checklist. Reference [38] is cited in the supplementary materials.

Author Contributions: J.C.: Conceptualization, methodology, literature search, data analysis, writing original draft preparation. J.T.F.: Supervision, methodological validation, critical review and editing. H.R.: Review and editing, validation of technical content. A.J.M.C.: Review and editing, validation of technical content. A.C.: Contribution to aviation context analysis and review. P.G.: Contribution to military aviation perspective and validation of operational aspects. J.F.: Contribution to domain expertise, review and validation. All authors have read and agreed to the published version of the manuscript.

Funding: This work was supported by the Portuguese Foundation for Science and Technology (FCT) under Projects UID/4131/2025 (https://doi.org/10.54499/UID/04131/2025) and UID/PRR/4131/2025 (https://doi.org/10.54499/UID/PRR/04131/2025).

Institutional Review Board Statement: Not applicable.

Informed Consent Statement: Not applicable.

Data Availability Statement: This study is based on a systematic literature review of previously published and publicly available scientific articles. No new datasets were generated or analysed during the current study. All data supporting the findings of this review are available within the articles and its referenced sources.

Acknowledgments: The authors appreciate the technical and infrastructural support provided by the Portuguese Air Force, which was essential in helping with this literature review. The authors also extend their sincere appreciation to the RCM2+ (Research Centre for Asset Management and Systems Engineering) and CISE—Electromechatronic Systems Research Centre, for their continued support in fostering research excellence in predictive maintenance.

Conflicts of Interest: The authors declare no conflicts of interest.

Abbreviations

The following abbreviations are used in this manuscript:

AE	Autoencoder
AI	Artificial Intelligence
ANN	Artificial Neural Network
ARIMA	Autoregressive Integrated Moving Average
CBM	Condition-Based Maintenance
CNN	Convolutional Neural Network
DT	Digital Twin
GAN	Generative Adversarial Network
GCN	Graph Convolutional Network
GRU	Gated Recurrent Unit
IoT	Internet of Things

LSTM	Long Short-Term Memory
MBSE	Model-Based Systems Engineering
ML	Machine Learning
MLP	Multilayer Perceptron
MRO	Maintenance, Repair, and Overhaul
PdM	Predictive Maintenance
PHM	Prognostics and Health Management
RCM	Reliability-Centered Maintenance
RNN	Recurrent Neural Network
RUL	Remaining Useful Life
SHAP	SHapley Additive exPlanations
SVM	Support Vector Machine
UAV	Unmanned Aerial Vehicle
XAI	Explainable Artificial Intelligence

References

1. Stanton, I.; Munir, K.; Ikram, A.; El-Bakry, M. Predictive maintenance analytics and implementation for aircraft: Challenges and opportunities. *Syst. Eng.* **2023**, *26*, 216–237. [CrossRef]
2. Teubert, C.; Pohya, A.A.; Gorospe, G. An Analysis of Barriers Preventing the Widespread Adoption of Predictive and Prescriptive Maintenance in Aviation. 2023. Available online: https://ntrs.nasa.gov/citations/20230000841 (accessed on 28 January 2025)
3. Meissner, R.; Rahn, A.; Wicke, K. Developing prescriptive maintenance strategies in the aviation industry based on a discrete-event simulation framework for post-prognostics decision making. *Reliab. Eng. Syst. Saf.* **2021**, *214*, 107812. [CrossRef]
4. Tseremoglou, I.; Santos, B.F. Condition-Based Maintenance scheduling of an aircraft fleet under partial observability: A Deep Reinforcement Learning approach. *Reliab. Eng. Syst. Saf.* **2024**, *241*, 109582. [CrossRef]
5. Orošnjak, M.; Saretzky, F.; Kedziora, S. Prescriptive Maintenance: A Systematic Literature Review and Exploratory Meta-Synthesis. *Appl. Sci.* **2025**, *15*, 8507. [CrossRef]
6. Kabashkin, I.; Fedorov, R.; Perekrestov, V. Decision-Making Framework for Aviation Safety in Predictive Maintenance Strategies. *Appl. Sci.* **2025**, *15*, 1626. [CrossRef]
7. Li, Y.F.; Wang, H.; Sun, M. ChatGPT-like large-scale foundation models for prognostics and health management: A survey and roadmaps. *Reliab. Eng. Syst. Saf.* **2024**, *243*, 109850. [CrossRef]
8. Costa, J.P.; Farinha, J.T.; Mendes, M.; Estima, J.O. AI-Driven Belt Failure Prediction and Prescriptive Maintenance with Motor Current Signature Analysis. *Appl. Sci.* **2025**, *15*, 6947. [CrossRef]
9. Vanderschueren, T.; Boute, R.; Verdonck, T.; Baesens, B.; Verbeke, W. Prescriptive maintenance with causal machine learning. *arXiv* **2022**, arXiv:2206.01562. [CrossRef]
10. Yang, H.; Desell, T. A Large-Scale Annotated Multivariate Time Series Aviation Maintenance Dataset from the NGAFID. *arXiv* **2022**, arXiv:2210.07317. [CrossRef]
11. Ziyad, M.; Tjandra, K.; Sugihartanto, M.F.; Arief, M. An Optimized and Safety-aware Maintenance Framework: A Case Study on Aircraft Engine. In Proceedings of the IEEE Conference on Intelligent Transportation Systems ITSC, Macau, China, 12–18 October 2022. [CrossRef]
12. Nanyonga, A.; Wasswa, H.; Joiner, K.; Turhan, U.; Wild, G. Explainable Supervised Learning Models for Aviation Predictions in Australia. *Aerospace* **2025**, *12*, 223. [CrossRef]
13. Zhao, Y.; Yang, J.; Wang, W.; Yang, H.; Niyato, D. TranDRL: A Transformer-Driven Deep Reinforcement Learning Enabled Prescriptive Maintenance Framework. *IEEE Internet Things J.* **2024**, *11*, 35432–35444. [CrossRef]
14. van Dinter, R.; Tekinerdogan, B.; Catal, C. Predictive maintenance using digital twins: A systematic literature review. *Inf. Softw. Technol.* **2022**, *151*, 107008. [CrossRef]
15. Apostolidis, A.; Stamoulis, K.P. An AI-based Digital Twin Case Study in the MRO Sector. *IFAC-PapersOnLine* **2020**, *56*, 55–62. [CrossRef]
16. Yang, H.; LaBella, A.; Desell, T. Predictive Maintenance for General Aviation Using Convolutional Transformers. In Proceedings of the AAAI Conference on Artificial Intelligence, Online, 22 February–1 March 2022; Volume 36, pp. 12636–12642. [CrossRef]
17. Jammal, P.; Pinon Fischer, O.; Mavris, D.N.; Wagner, G. Advancing Aviation Safety Through Predictive Maintenance: A Machine Learning Approach for Carbon Brake Wear Severity Classification. *Aerospace* **2025**, *12*, 602. [CrossRef]
18. Al Hasib, A.; Rahman, A.; Khabir, M.; Shawon, M.T.R. An Interpretable Systematic Review of Machine Learning Models for Predictive Maintenance of Aircraft Engine. In *Data Driven Applications for Industry 4.0 and Beyond*; Taylor & Francis Group: Abingdon, UK, 2025. [CrossRef]

19. Martins, A.; Fonseca, I.; Farinha, J.T.; Reis, J.; Cardoso, A.J.M. Maintenance Prediction through Sensing Using Hidden Markov Models—A Case Study. *Appl. Sci.* **2021**, *11*, 7685. [CrossRef]
20. Eddarhri, M.; Adib, J.; Hain, M.; Marzak, A. Towards predictive maintenance: The case of the aeronautical industry. *Procedia Comput. Sci.* **2022**, *203*, 769–774. [CrossRef]
21. Fu, X.; Luo, H.; Zhong, S.; Lin, L. Aircraft engine fault detection based on grouped convolutional denoising autoencoders. *Chin. J. Aeronaut.* **2019**, *32*, 296–307. [CrossRef]
22. Ganie, A.B.; Singh, E.J. Predictive Maintenance of Aircrafts on Large Scale Industrial Units Using Machine Learning Algorithms. *Int. J. Res. Appl. Sci. Eng. Technol.* **2022**, *10*, 1343–1352. [CrossRef]
23. Bhat, O.J.; Kukana, P. Predictive Maintenance of Aircrafts on Large Scale Industrial Units Using Machine Learning Algorithms. *J. Propuls. Technol.* **2024**, *45*, 636–643. [CrossRef]
24. Patibandla, K.R. Predictive Maintenance in Aviation using Artificial Intelligence. *J. Artif. Intell. Gen. Sci. (JAIGS)* **2024**, *4*, 325–333. [CrossRef]
25. Pose, C.; Giribet, J.; Torre, G. Propeller damage detection, classification and estimation in multirotor vehicles. *arXiv* **2024**, arXiv:2410.05447. [CrossRef]
26. e Lima, N.A.F.S. Development of an Aircraft Health Monitoring Program for Predictive Maintenance. Master's Thesis, Instituto Superior Técnico de Lisboa, Lisboa, Portugal, 2017.
27. Cha, G.; Park, J.; Moon, I. Military aircraft flight and maintenance planning model considering heterogeneous maintenance tasks. *Reliab. Eng. Syst. Saf.* **2023**, *239*, 109497. [CrossRef]
28. Thompson, N.; Blond, K.; Clark, A.; Shearin, M. MBSE Applications to Optimize Predictive Maintenance Scheduling in Military Aviation. In Proceedings of the 2023 IEEE Aerospace Conference, Big Sky, MT, USA, 4–11 March 2023; pp. 1–9. [CrossRef]
29. O'Neil, R.; Khatab, A.; Diallo, C. Optimizing predictive maintenance and mission assignment to enhance fleet readiness under uncertainty. *Auton. Intell. Syst.* **2025**, *5*, 17. [CrossRef]
30. Khan, Z.U.; Nasim, B.; Rasheed, Z. Generative AI based Predictive Maintenance in Aviation: A Systematic Literature Review. *CEAS Aeronaut. J.* **2024**, *16*, 537–555. [CrossRef]
31. Grehan, J.; Ignatyev, D.; Zolotas, A. Fault Detection in Aircraft Flight Control Actuators Using Support Vector Machines. *Machines* **2023**, *11*, 211. [CrossRef]
32. Li, Z.; Ma, J.; Fan, R.; Zhao, Y.; Ai, J.; Dong, Y. Aircraft Sensor Fault Diagnosis Based on GraphSage and Attention Mechanism. *Sensors* **2025**, *25*, 809. [CrossRef]
33. Dangut, M.D.; Jennions, I.K.; King, S.; Skaf, Z. A rare failure detection model for aircraft predictive maintenance using a deep hybrid learning approach. *Neural Comput. Appl.* **2023**, *35*, 2991–3009. [CrossRef]
34. Pebrianti, D.; Khalani, Z.; Bayuaji, L. Predictive Maintenance in Aerospace Industry Using Convolutional Neural Network. In Proceedings of the 2024 9th International Conference on Mechatronics Engineering (ICOM), Kuala Lumpur, Malaysia, 13–14 August 2024; pp. 157–162. [CrossRef]
35. Basora, L.; Bry, P.; Olive, X.; Freeman, F. Aircraft Fleet Health Monitoring with Anomaly Detection Techniques. *Aerospace* **2021**, *8*, 103. [CrossRef]
36. Heim, S.; Clemens, J.; Steck, J.E.; Basic, C.; Timmons, D.; Zwiener, K. Predictive Maintenance on Aircraft and Applications with Digital Twin. In Proceedings of the 2020 IEEE International Conference on Big Data (Big Data), Atlanta, GA, USA, 10–13 December 2020; pp. 4122–4127. [CrossRef]
37. Pater, I.; Mitici, M. Predictive maintenance for multi-component systems of repairables with Remaining-Useful-Life prognostics and a limited stock of spare components. *Reliab. Eng. Syst. Saf.* **2021**, *214*, 107761. [CrossRef]
38. Page, M.J.; McKenzie, J.E.; Bossuyt, P.M.; Boutron, I.; Hoffmann, T.C.; Mulrow, C.D.; Shamseer, L.; Tetzlaff, J.M.; Akl, E.A.; Brennan, S.E.; et al. The PRISMA 2020 statement: An updated guideline for reporting systematic reviews. *BMJ* **2021**, *372*, n71. [CrossRef]